DAY 5 · CLOUD LANDING ZONES

Week 1 in a new cloud decides the next 5 years.



DEJI AJISHE · DevOps | Multi-Cloud

The 6 foundations →

CLOUD LANDING ZONES

Before you deploy to three clouds, design these 6 things first

A landing zone is the foundation every workload inherits. Retrofitting it later is the expensive path.

1 Account structure

Separate prod, non-prod, security and logging.

Organizations + OUs · Management groups · Folders + projects

2 Identity

Central SSO, group-based roles, no shared admin users.

IAM Identity Center · Entra ID · Cloud Identity

3 Network

Planned IP ranges, hub-and-spoke, controlled egress.

Transit Gateway · Hub VNet / Virtual WAN · Shared VPC

4 Logging

Org-wide audit logs into a locked-down archive.

Org CloudTrail · Diagnostic settings · Aggregated sinks

5 Security guardrails

Preventive policies that apply before anyone deploys.

SCPs · Azure Policy · Organization Policy

6 Governance

Tagging, budgets and ownership from day one.

Tag policies · Cost Management · Labels + budgets

01 · THE HOOK

The first week in a new cloud shapes the next five years

Most painful cloud clean-ups trace back to decisions nobody made on purpose.

Everything in one account. Admin rights shared "for now." Audit logs that the same admins can delete.

A **landing zone** is the foundation every future workload inherits. Design it before the first workload arrives.

02 · THE PROBLEM

What "just start deploying" costs later

- → **No isolation:** dev mistakes and prod incidents share the same blast radius
- → **Shared admin access:** no clear answer to "who can change production?"
- → **Deletable logs:** the people being audited can erase the audit trail
- → **No cost attribution:** the bill grows, and nobody knows which team owns what
- → **Retrofitting is hard:** moving live workloads between accounts, subscriptions or projects takes real effort

Separate by purpose, then by environment

```
AWS Organization
├── Security OU       # log archive, security tooling
├── Infrastructure OU # shared networking
├── Workloads OU → prod / non-prod
└── Sandbox OU        # looser guardrails, budgets

Azure: tenant root management group
├── Platform  # identity, management, connectivity
├── Landing zones → corp / online
└── Sandbox

GCP Organization
├── common / bootstrap folders
└── prod / nonprod / dev folders → projects
```

Each cloud's reference design follows the same idea: platform first, workloads underneath.

The 6 foundations, cloud by cloud

	AWS	Azure	GCP
Structure	Organizations + OUs	Management groups	Folders + projects
Identity	IAM Identity Center	Entra ID + PIM	Cloud Identity
Network	Transit Gateway	Hub VNet / Virtual WAN	Shared VPC
Logging	Organization CloudTrail	Diagnostic settings → Log Analytics	Aggregated log sinks
Guardrails	SCPs	Azure Policy	Organization Policy
Governance	Tag policies, budgets	Tags, Cost Management	Labels, budgets

> **Don't start from a blank page:** AWS Control Tower, Azure Landing Zones and Google's enterprise foundations blueprint are all good starting points.

Guardrails as code

AWS SCP: nobody can switch off the audit trail

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Sid": "ProtectAuditTrail",
    "Effect": "Deny",
    "Action": [
      "cloudtrail:StopLogging",
      "cloudtrail:DeleteTrail"
    ],
    "Resource": "*"
  }]
}
```

→ **Azure:** assign the built-in **Allowed locations** policy at a management group

→ **GCP:** enforce `iam.disableServiceAccountKeyCreation` at the organization

05 · THE TRADE-OFFS

Guardrails vs developer speed

→ **Too loose:** drift, shadow admin access and surprise bills

→ **Too strict:** teams wait on tickets and find workarounds

→ **Balance:** strict in prod, looser in a budget-capped sandbox

→ **Automate account/project creation** so the safe path is also the fast path

Principle: preventive guardrails for things that must never happen, detective controls for everything else.

06 · THE TAKEAWAY

Landing zone checklist

- ☐ Separate accounts/subscriptions/projects for prod, non-prod, security and logging
- ☐ Central SSO with group-based roles; break-glass access documented
- ☐ IP plan and hub-and-spoke network decided up front
- ☐ Org-wide audit logs in a separate, locked-down archive
- ☐ Preventive guardrails applied at the org/root level
- ☐ Mandatory tags/labels and budgets with alerts
- ☐ Account/project creation automated

FURTHER READING

Go deeper: official references

AWS: What is AWS Control Tower?
docs.aws.amazon.com/controltower/latest/userguide/what-is-control-tower.html

AWS: Service control policies (SCPs)
docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_scps.ht

Microsoft Learn: Azure landing zones
learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/

Google Cloud: Enterprise foundations blueprint
cloud.google.com/architecture/security-foundations

Google Cloud: Organization Policy Service
cloud.google.com/resource-manager/docs/organization-policy/overview

Docs change. Always check the current version before you design.

07 · YOUR TURN

What is the first control you establish in a new cloud environment?

Share yours in the comments. I'm curious which one engineers put first.

Follow Deji Ajishe

for the rest of this 30-part multi-cloud DevOps series

♻️ Repost or share this with an engineer on your team

linkedin.com/in/deji-ajishe-64a35722a